

Saranya Vijayakumar

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Education

Carnegie Mellon University

PITTSBURGH

Ph.D., Computer Science

Expected graduation 2026

M.S., Computer Science Research

2025 via Ph.D program

Advisors: Professors Christos Faloutsos & Matt Fredrikson

Harvard University

CAMBRIDGE

A.B., Joint Concentration in Computer Science & Government

2018

Thesis: "Interpretability Through Interrogation: Fairness in the Context of Criminal Sentencing"

Bachelor's Advisors: Professors Cynthia Dwork & Jim Waldo

Conference Publications

[1] **Neuro-Symbolic Prover-Verification for In-Context Boolean Function Learning**

Saranya Vijayakumar, Christos Faloutsos, Matt Fredrikson

Upcoming

2026

[2] **Bipartite Graph Feature Engineering for Credit Card Fraud Detection at Scale**

Saranya Vijayakumar, Christos Faloutsos, Robson Cordeiro

Upcoming

2026

[3] **Reputation-Weighted Gossip Enables Robust Deception Detection in Adversarial Multi-Agent Systems**

Saranya Vijayakumar, Koichi Onoue

Upcoming

2026

[4] **Scalable Oversight via Prover Multi-Verifier Games**

Saranya Vijayakumar, Erik Miehl, Karthikeyan Ramamurthy, Dennis Wei, Djallel Bouneffouf, Murray Campbell

Upcoming

2026

[5] **Evaluating LLM-Supported Malware Evasion: A Red Team Benchmark for Code Obfuscation and Antivirus Bypass**

Saranya Vijayakumar, Christos Faloutsos, Matt Fredrikson

Upcoming

2026

[6] **Leveraging Large Language Models for Enhanced Membership Inference and Reidentification in Topics API Analyses**

Saranya Vijayakumar, Norman Sadeh

ICISSP (Best Student Paper Award)

Marbella 2026

[7] **Prototype-Integrated Representation Learning for Novelty Detection**

Saranya Vijayakumar, Christos Faloutsos, Matt Fredrikson

IEEE Trust, Security and Privacy in Computing and Communications (TrustCom)

Hong Kong 2025

[8] **AICodeDetect: A Pipeline for Systematic Detection and Analysis of AI-Generated Code**

Saranya Vijayakumar, Philip Negrin, Christos Faloutsos

IEEE Mathematics and Computers in Sciences and Industry (MCSI)

Rhodes 2025

[9] **Mechanistically Interpreting a Transformer-based 2-SAT Solver: An Axiomatic Approach**

Nils Palumbo, Ravi Mangal, Zifan Wang, Saranya Vijayakumar, Corina Pasareanu, Somesh Jha

International Conference on Machine Learning (ICML)

Vancouver 2025

[10] **Aligned LLMs Are Not Aligned Browser Agents**

Priyanshu Kumar, Saranya Vijayakumar, Elaine Lau, Tu Trinh, Zifan Wang, Matt Fredrikson

The International Conference on Learning Representations (ICLR) (Paper)

Singapore 2025

[11] **Grounding Neural Inference with Satisfiability Modulo Theories**

Saranya Vijayakumar, Zifan Wang, Kaiji Lu, Vijay Ganesh, Somesh Jha, Matt Fredrikson

NeurIPS (Spotlight) (Talk)

Vancouver 2023

[12] **CallMine: Fraud Detection and Visualization of Million-Scale Call Graphs**

Mirela Cazzolato, Saranya Vijayakumar, Meng-Chieh Lee, Namyoung Park, Catalina Vajiac, Christos Faloutsos

The Conference on Information and Knowledge Management (CIKM)

Birmingham 2023

Workshop Publications and Conference Contributions

- [1] **Through the Lens of LLMs: Unveiling Differential Privacy Challenges**
USENIX Conference on Privacy Engineering Practice and Respect (PEPR) Santa Clara 2024
- [2] **Anomaly Detection and Visualization of Large-Scale Call Graphs**
AAAI-23 Demonstrations Program Washington DC 2023
- [3] **TgraphSpot: Fast and Effective Anomaly Detection for Time-Evolving Graphs**
2022 IEEE International Conference on Big Data Industry and Government Program Osaka 2022
- [4] **Interpretability Through Interrogation: Fairness in the Context of Criminal Sentencing** 2018
- [5] **Algorithmic Decision-Making** Harvard Political Review, 2017
- [6] **A Worldwide Survey of Encryption Products**
Bruce Schneier, Kathleen Seidel, and Saranya Vijayakumar. SSRN, 2015

Invited Talks

- [1] 17-416/17-716, AI Governance (Masters/PhD level) CARNEGIE MELLON UNIVERSITY
Guest Lecture on LLM Security and Alignment; post-training techniques (SFT, DPO, RLHF/RLAIF, preference learning, etc. Spring 2025
- [2] 17-331/17-631, Information Security, Privacy, Public Policy CARNEGIE MELLON UNIVERSITY
Guest Lecture on Vulnerabilities of ML Fall 2024
- [3] 17-416/17-716, AI Governance (Masters/PhD level) CARNEGIE MELLON UNIVERSITY
Guest Lecture on ML Security and Privacy Spring 2024
- [4] 17-331/17-631, Information Security, Privacy, Public Policy CARNEGIE MELLON UNIVERSITY
Guest Lecture on ML Security and Adversarial Robustness Fall 2023
- [5] **Dagstuhl Seminar: Machine Learning and Logical Reasoning: The New Frontier** GERMANY 2022
- [6] **CRA-WP Grad Cohort 2022** NEW ORLEANS 2022
- [7] **Cylab Partners Conference** PITTSBURGH 2022
- [8] **Alumni Committee for Harvard Women in Computer Science** 2022

Selected Fellowships and Awards

- [1] **ML and Systems Rising Stars Award, MLCommons** CALIFORNIA, 2026
- [2] **Best Student Paper Award, ICISSP** MARBELLA, 2026
- [3] **Best Poster Award, NDSEG** NEW ORLEANS 2024
- [4] **GFSD Program** NSA
- [5] **CRA-W Grad Cohort for Women** NEW ORLEANS 2022
- [6] **National Defense Science & Engineering Graduate Fellowship Program (NDSEG)** ARMY RESEARCH OFFICE, 2022 – 2025
- [7] **Tech in the World Fellow, Partners in Health** LIMA 2018
- [8] **The Ernst Kitzinger Prize, Lowell House** HARVARD UNIVERSITY, 2018
- [9] **Microsoft Scholarship, Grace Hopper Celebration of Women in Computing** ORLANDO 2017
- [10] **Director's Internship, Harvard Kennedy School Institute of Politics** NEW YORK 2015

Teaching Experience

- 17-331/631, Information, Security, Privacy & Policy (Masters level) CARNEGIE MELLON UNIVERSITY
Teaching Assistant Fall 2023
Created homework assignments, graded assignments, and held office hours. Gave a lecture on ML/security. Course included applied cryptography, authentication & security protocols, web & network attacks, and ML security & privacy.
- 15-294, Rapid Prototyping (undergraduate level) CARNEGIE MELLON UNIVERSITY
Teaching Assistant Spring 2023
Taught lecture, graded assignments, and redesigned the syllabus and course schedule to accommodate interactive learning and new assignments. Course focused on SolidWorks.

15-394, Intermediate Rapid Prototyping

CARNEGIE MELLON UNIVERSITY

Teaching Assistant

Spring 2023

Taught lecture, graded assignments, and redesigned homework assignment for students to design automata in SolidWorks with linear bushing, rotational motion, and rendering/motion analysis components. Course focused on Rhino, Grasshopper, and Kangaroo 2 physics-based simulation.

Future Faculty Program

CARNEGIE MELLON UNIVERSITY

Participant

2021 – 2023

Eberly Center for Teaching Excellence & Educational Innovation. Participated in seminars aimed at helping graduate students develop and document their teaching skills in preparation for a faculty career. Completed a lesson plan review and teaching observation with Eberly experts; redesigned Rapid Prototyping syllabus; completed a teaching philosophy project

Industry Collaborations

Fujitsu/Agentic AI

PITTSBURGH

Intern

January – May 2026

- Studying red-teaming of agentic AI with Koichi Onoue and Roshni Kaushik. Architecting, building, and deploying LLM-powered and agent-based systems end-to-end

IBM/Trustworthy AI

YORKTOWN HEIGHTS

Intern

May – August 2025

- Game theoretic extension of prover-verifier games for legibility using multiple specialized agentic verifiers. Studied under Erik Miehl and Karthikeyan Ramamurthy. Emphasis on foundational LLMs, agentic AI, reinforcement learning (RLHF).

Inria/Proof techniques for security protocols (PESTO)

NANCY

Visiting Scholar

October – November 2024

- Formal verification project: Studying the security properties and transcript consistency of a secure messaging platform used by the French government. Studied under Charlie Jacomme and Steve Kremer.

Mobileum/Adaptive, Intelligent and Distributed Assurance Platform (AIDA)

BRAGA

Collaborator

2021 – 2026

- Collaborated with Mobileum, a global provider of telecom analytics solutions, on industry-scale fraud detection research. Mobileum offers risk management, roaming, and network analytics to over 900 telecom operators worldwide.
- Analyzed real-world call graph data to develop scalable anomaly detection techniques.
- Published multiple peer-reviewed publications and demos, including deployment-ready systems for detecting telecom fraud on million-scale graphs.

Goldman Sachs/Algorithmic Trading (GSET)

NEW YORK

Data Scientist, Electronic Trading

2018 – 2021

- Covered quantitative hedge funds and asset managers in a client-facing data science role.
- Performed trade cost analyses using Python, Slang (Goldman's proprietary language), SQL, and KDB Q and communicated algorithmic recommendations to stakeholders.
- Designed and implemented experiments with the software engineering team & strategized on new features and methodologies.
- Published research pieces sent to over one thousand clients, focusing on market microstructure and electronic trading statistics.

Beto O'Rourke for U.S. Senate

AUSTIN

Data Scientist, Distributed Organizing

Summer 2018

- Collaborated with the data team and campaign director to create Python models predicting voter turnout and support.
- Presented my findings on persuasion tactics and priority counties for grassroots organizing to the chief of staff.
- Strategized student turnout and started grassroots offices around Texas.
- Canvassed in San Antonio with the Field Director to organize and fundraise.

Booz Allen Hamilton

VIRGINIA SQUARE, HERNDON & BOSTON

Cybersecurity Intern

Summer 2017

- Worked on autonomous swarming behavior in team of six.

- Created the functionality for semi-autonomous navigation of the ground robots in ROS and using Python and C++.
- Investigated and created prevention methods against security threats by creating a proof-of-work demonstrating how a potential hacker could use GPS spoofing to override a military-grade GPS-enabled robot. Implemented PCA for GPS anomaly detection.

Digital Solutions Intern

Winter 2017

- Collaborated with intern team to perform impact analysis on the MBTA using data provided by the MBTA to evaluate pricing strategies based on revenue and equity.
- Studied the fairness of public transit fares in Boston by examining surge pricing, subsidies to low-income individuals and students, and distance-based fares.

Conference Service

Program Committee, Foundations of Agentic Systems Theory	NeurIPS 2025
NeurIPS, Reviewer	NeurIPS 2025
ICML, Reviewer	ICML 2025
ICLR, Reviewer	ICLR 2025
KDD, Reviewer	KDD 2025
ICLR, Reviewer	ICLR 2024
NeurIPS, Reviewer	NeurIPS 2023 Workshop: New Frontiers in Graph Learning
NeurIPS, Reviewer	NeurIPS 2024
Peer Reviewer	Georgetown Center for Security and Emerging Technology (CSET), 2024

Selected Service

Women in CSD, Founder	CARNEGIE MELLON UNIVERSITY, 2022 – PRESENT
• Organizer of a weekly lunch and other programming for over 90 women and non-binary members of Computer Science Department and broader School of Computer Science.	
Introductory Course, Organizer	CARNEGIE MELLON UNIVERSITY, SUMMER 2022
• Organized Introductory Course events, which introduces new Ph.D. students to the department.	
Alumni Association Executive Committee, Member	RIVERDALE COUNTRY SCHOOL, 2022 - PRESENT
• Sustain loyalty and enthusiasm among peers by developing programs, initiatives, and events that promote the general welfare of the school and by encouraging alumni participation in activities and philanthropic support. • Class Correspondent (2014 - present): Solicit class notes & serve as part of Reunion Committee	
Harvard University, Member	BOSTON & NEW YORK, 2018 - PRESENT
• Schools & Scholarships Committee (2018 - Present): interview Harvard College applicants yearly • Participation chair for class of 2018 fifth year reunion (2023)	
Cyber Defense Club, Finance & Communications Chair	HARVARD UNIVERSITY, 2017 – 2018
• Qualified for the New England regional finals of the National Collegiate Cyber Defense Competition. • Led weekly meetings.	
Girls Who Code, Leader	HARVARD UNIVERSITY, 2016 - 2017
• Wrote the curriculum and led classes for 20 middle school girls, coordinated female Harvard mentors. • Collaborated with Harvard Kennedy School and Business School students to make Girls Who Code more inclusive to girls in different parts of Boston. • Worked with the Harvard EdLabs to host programs in Allston, MA.	
Digital Literacy Project, Education & Community Outreach Chairs	HARVARD UNIVERSITY, 2014 - 2017
• Taught computer science weekly at underserved schools in Boston. • Expanded the curriculum to create second course in HTML and CSS and created partnerships with Boston public schools.	

Skills

Technical specialties: Java, Python, C, R, Tensorflow, Sklearn, ROS, HTML+CSS, PHP, Swift, Git, Parse, Open-MRS, Stata, $\LaTeX$. Linux administration skills: bash, Apache, MySQL, VMware, & KDB Q.

Natural languages: English, Tamil, Spanish (*working proficiency*), Japanese (*limited working proficiency*).
